

REV2 REFERENCE / REDESIGN IN PROGRESS

22 September 2026: the current print order is N1 nose only.

Use Manufacturing/Nose_Print_N1/A59_Nose_N1_mm.stl and Nose_N1_Print_Guide.pdf. Quantity one, millimetres, 100%.

The following pages retain the old Rev2 design for reference. They are not the current full-aircraft manufacturing instructions. Their original page numbers remain unchanged; this notice is an additional page.

Superseded for the new design

Landing gear and steering system; the old 42-piece print order; A59 cover screw holes, columns, spacers and shaped washers; the large nose pitot-access notch. Do not fabricate these from the old instructions.

N1 restores the original upper nose shell. It uses a proposed simple ledge/tape attachment and a probe passage drilled to actual hardware during assembly. It retains the original rear mating envelope.

The rest of the aircraft will be simplified around the plywood skeleton and foamboard. Remaining mounts, control construction, belly protection, installation details and mass/CG estimates still require that redesign. Do not use the old mass, CG or gear clearance numbers for the new configuration.

Existing cutting templates are retained until a reuse/change list is complete. The N1 release is a nose-only handoff, not a completed redesign of every file in this manual.

Parts and materials

Quantities are for one aircraft unless marked as ground equipment. The process coupons and forming tools are not installed parts. Use the CSV for file links and detailed fastening notes.

Plywood

ID	Part	Qty	Material / specification
A01	Rear joiner support	2	5.000 mm aircraft plywood
A04	Aft fuselage former	1	5.000 mm aircraft plywood
A05	Front wing spar	2	5.000 mm aircraft plywood
A07	Rear wing spar	2	5.000 mm aircraft plywood
A10	Outer aileron end plate	2	3.000 mm aircraft plywood
A13	Forward longitudinal former	1	5.000 mm aircraft plywood
A16	Middle fuselage former	1	5.000 mm aircraft plywood
A18	Tail bulkhead	1	2.500 mm aircraft plywood
A24	Middle wing rib	2	5.000 mm aircraft plywood
A25	Front gear bulkhead	1	5.000 mm aircraft plywood
A26	Motor mount	1	5.000 mm aircraft plywood
A34	Forward fuselage former	1	5.000 mm aircraft plywood
A35	Inner wing rib	2	5.000 mm aircraft plywood
A43	Aft longitudinal former	1	5.000 mm aircraft plywood
A44	Outer wing rib	2	4.990 mm aircraft plywood
A47	Main gear base	1	5.000 mm aircraft plywood
A48	Rear gear bulkhead	1	5.000 mm aircraft plywood
A53	Inner aileron end plate	2	3.000 mm aircraft plywood
A54	Central forward former	1	5.000 mm aircraft plywood
A62F	Front nose spine segment	1	5.000 mm aircraft plywood
A62R	Rear spine segment	1	5.000 mm aircraft plywood
A63	Nose cheek doublers (A63/A64)	2	2.500 mm aircraft plywood
G03	Main gear mounting doubler	1	3.000 mm aircraft plywood
H05B	Control horn backing plate	4	2.500 mm aircraft plywood
I01	Adjustable battery tray	1	3.000 mm aircraft plywood

Printed parts

ID	Part	Qty	Material / specification
A11	Left aileron	1	LW-PLA preferred; PLA+ mass must be budgeted
A21	Right aileron	1	LW-PLA preferred; PLA+ mass must be budgeted
A23	Left aileron hinge	1	LW-PLA preferred; PLA+ mass must be budgeted
A36	Right aileron hinge	1	LW-PLA preferred; PLA+ mass must be budgeted
A45	Right wing root	1	PLA+ or PETG; functional coupon required
A55	Left wing root	1	PLA+ or PETG; functional coupon required
I04	FC Bridge	1	qualified PETG / PLA+

ID	Part	Qty	Material / specification
I05-1	Nose Cover Column	1	qualified PETG / PLA+
I05-2	Nose Cover Column	1	qualified PETG / PLA+
I05-3	Nose Cover Column	1	qualified PETG / PLA+
I05-4	Nose Cover Column	1	qualified PETG / PLA+
I07	FC Service Hatch	1	original shell print material
I08-1	Hatch Support Tab	1	qualified PETG / PLA+
I08-2	Hatch Support Tab	1	qualified PETG / PLA+
I13	Airspeed Board Carrier	1	PETG / PLA+
A37	Nose gear mount	1	qualified PLA+ or PETG
A09	Nose steering bearing	1	qualified PLA+ or PETG
I02	Battery Corner Stop	2	qualified PETG / PLA+
I17-1L	Battery Hanger	1	qualified PETG / PLA+
I17-1R	Battery Hanger	1	qualified PETG / PLA+
I17-2L	Battery Hanger	1	qualified PETG / PLA+
I17-2R	Battery Hanger	1	qualified PETG / PLA+
I00	Nose Cover Spacer 6mm	4	nylon or qualified print material
I10	Nose Receiver Carrier	1	qualified PETG / PLA+
A06	Centre fairing	1	original shell print material
I18	Battery Access Hatch	1	original shell print material
I19-1	Battery Hatch Tab	1	qualified PETG / PLA+
I19-2	Battery Hatch Tab	1	qualified PETG / PLA+
I12	Forward ESC Carrier	1	qualified PETG / PLA+
I11	Aft Telemetry Carrier	1	qualified PETG / PLA+
A12	Tail fairing	1	original shell print material
A59	Nose shell	1	PLA+ / nylon fittings
I09	Internal GPS Bridge	1	PLA+ / nylon fittings
I14	Slim Pitot Carrier	1	PLA+ / nonmagnetic fittings
I06-1	Compact Cover Washer	1	PLA+ / nonmagnetic fittings
I06-2	Compact Cover Washer	1	PLA+ / nonmagnetic fittings
I06-3	Compact Cover Washer	1	PLA+ / nonmagnetic fittings
I06-4	Compact Cover Washer	1	PLA+ / nonmagnetic fittings

Foam patterns

ID	Part	Qty	Material / specification
F01	Left inner wing	1	5 mm FliteBoard Pro
F02	Right inner wing	1	5 mm FliteBoard Pro
F03	Right outer wing	1	5 mm FliteBoard Pro
F04	Left outer wing	1	5 mm FliteBoard Pro
F05	Right fixed tail - main	1	5 mm FliteBoard Pro
F06	Left fixed tail - main	1	5 mm FliteBoard Pro

ID	Part	Qty	Material / specification
F07	Right fixed tail - root collar	1	5 mm FliteBoard Pro
F08	Left fixed tail - root collar	1	5 mm FliteBoard Pro
F09	Forward lower fuselage - left curve	1	5 mm FliteBoard Pro
F10	Forward lower fuselage - right curve	1	5 mm FliteBoard Pro
F11	Middle lower fuselage - right curve	1	5 mm FliteBoard Pro
F12	Middle lower fuselage - left curve	1	5 mm FliteBoard Pro
F13	Forward lower fuselage - left lip	1	5 mm FliteBoard Pro
F14	Forward lower fuselage - bottom strip	1	5 mm FliteBoard Pro
F15	Forward lower fuselage - right lip	1	5 mm FliteBoard Pro
F16	Middle lower fuselage - left lip	1	5 mm FliteBoard Pro
F17	Middle lower fuselage - right lip	1	5 mm FliteBoard Pro
F18	Middle lower fuselage - bottom strip	1	5 mm FliteBoard Pro
F19	Aft lower fuselage - left lip	1	5 mm FliteBoard Pro
F20	Aft lower fuselage - left curve	1	5 mm FliteBoard Pro
F21	Aft lower fuselage - right curve	1	5 mm FliteBoard Pro
F22	Aft lower fuselage - right lip	1	5 mm FliteBoard Pro
F23	Aft upper fuselage - left	1	5 mm FliteBoard Pro
F24	Aft upper fuselage - right	1	5 mm FliteBoard Pro
F25	Forward upper fuselage - right	1	5 mm FliteBoard Pro
F26	Forward upper fuselage - left	1	5 mm FliteBoard Pro
F27	Rear underside band 1 - forward	1	5 mm FliteBoard Pro
F28	Rear underside band 2	1	5 mm FliteBoard Pro
F29	Rear underside band 3	1	5 mm FliteBoard Pro
F30	Rear underside band 4 - aft	1	5 mm FliteBoard Pro
F31	Right tail control laminate blank	2	5 mm FliteBoard Pro
F32	Left tail control laminate blank	2	5 mm FliteBoard Pro
F33	Aft lower fuselage - bottom strip	1	5 mm FliteBoard Pro

Carbon tubes

ID	Part	Qty	Material / specification
T01	T01 front joiner	1	Carbon OD8/ID6 mm
T02	T02 right front sleeve	1	Carbon OD10/ID8.2 mm
T03	T03 left front sleeve	1	Carbon OD10/ID8.2 mm
T04	T04 rear joiner	1	Carbon OD8/ID6 mm

Hardware

ID	Part	Qty	Material / specification
H05	Control horns	4	Purchased lightweight RC horns
H07	Pushrod and clevis sets	5	Matched RC linkage

ID	Part	Qty	Material / specification
H08	Aileron pins	2	3 mm smooth rod; start 285 mm
G01	Main landing gear	1	20 x4 mm 6061-T6 aluminium
G02	Steerable nose gear	1	4 mm spring wire, shop-qualified forming
W01	Lightweight wheels	3	76.2 mm OD, 22 mm nominal hub width, 4 mm bore
H10	Electronics carrier fasteners	20	M2 x16 starting length, washers and locking nuts
H11	Airspeed carrier fasteners	6	Four M2 x16 foot bolts + two M2 PCB screws
H12	Battery hanger fasteners	8	Four M2 x16 foot bolts + four M2 finished 35-37 mm tray bolts
H13	Battery stop fasteners	4	M2 x12 starting length, 4-5 mm OD washers, locking nuts
H14	Nose cover fasteners	4	Nonmagnetic M2 brass/nylon: two x80 and two x60 starting lengths, washers/locking nuts
H15	Internal GPS fasteners	8	Four M2 x25 nylon bridge bolts + four M2 x14 nylon PCB bolts; washers and nuts
H16	FC stand-offs	4	M3 nylon 6 mm spacers, screws and supplied grommets
H17	Servo mounting fasteners	10	M2 through-bolts, broad washers and locking nuts
H18	Cable and sensor installation	1 set	Soft straps, tie bases, grommets, extension leads, heat-shrink, XPS-safe adhesive
H20	Wing root retainers	6	M2 x20 starting length; 12 washers and 6 locking nuts
H21	Nose bypass fasteners	4	Two M2 x40-45 inner bolts and two M2 x16 outer bolts; washers and locking nuts
H22	Nose bearing fasteners	4	M3 x16-20 starting length; washers and locking nuts
H23	Motor screws	4	M3 with washers; length from measured thread depth
H24	Control horn fasteners	8	M2 through-bolts, washers, locking nuts and crush-resistant spacer sleeves
H25	Pitot tab fastener	1	M2 x10 starting length, two washers and locking nut; two small anti-rotation ties
H01	Main gear bolts	2	M3 x20 starting length, broad washers and locking nuts
H02	Main wheel axles	2	M4 x40 starting length, smooth shank, spacers and locking nuts
H03	Nose arm and collars	1 arm + 4 collars	4 mm clamp steering arm and 4 mm shaft collars
H04	Battery straps	2	20 mm wide, at least 250 mm long, non-stretch hook-and-loop

Electronics

ID	Part	Qty	Material / specification
E01	Flight controller	1	Matek F405-WING V2
E02	GNSS and compass	1	Matek M9N-5883
E03	Airspeed kit	1	Matek ASPD-4525 with pitot and two tubes
E04	Receiver	1	FlySky FS-IA10B
E05	ESC	1	ReadytoSky 40 A, 2-4S, 5 V /3 A BEC

ID	Part	Qty	Material / specification
E06	Flight battery	1	GenX 4S 5200 mAh 40C; 138 x43 x33 mm, 512 g
E07	Motor	1	Emax ECO II 2807 1300 KV
E08	Propeller	1	Gemfan 9045CL, 9-inch three-blade, selected handedness
E09	Telemetry pair	1	SiK 433 MHz, user-specified 100 mW; only air radio flies
E10	Transmitter	1	FlySky FS-i6X; ground equipment
E11	Servos	5	TowerPro MG90S; nominal 13.4 g each